Sushil Thakur

Software Engineer – R&D | .NET 10 | Avalonia UI | WPF | ASP.NET Core | Desktop & Enterprise Applications

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Profile Summary

Software Engineer with 3+ years of experience in .NET technologies, specializing in Desktop Application Development, Enterprise Software Solutions, and Backend Engineering. Currently working in the Research & Development (R&D) team at CP Plus (Aditya Infotech Ltd), contributing to the development of Video Management System (VMS) products using .NET 10, AvaloniaUI, and WPF. Skilled in ASP.NET Core, C#, REST APIs, SQL Server, PostgreSQL, MVVM, and Clean Architecture, with hands-on experience in designing scalable, high-performance, and maintainable applications. Contributed to 27+ software projects across Smart City, Video Surveillance, IoT, and Enterprise domains, focusing on system performance, reliability, and user experience. Passionate about software architecture, desktop application development, and building modern cross-platform solutions while continuously enhancing expertise in system design, performance optimization, and emerging technologies.

Professional Key Competencies

Full-Stack Web Development

- **End-to-End Solutions:** Engineered scalable web applications leveraging C#, .NET 8/10+, and modern JavaScript frameworks to deliver high-performance user experiences.
- **Frontend Engineering:** Developed responsive, component-based UIs using Angular and React, ensuring seamless cross-browser compatibility and optimized state management.

Tech Stack

- **Backend & Frameworks: .**NET 8/10+, ASP.NET Core, MVC, REST API, CQRS, Minimal API
- **Desktop Application Development:** WPF, Avalonia UI, MVVM Architecture, Dependency Injection (DI), Custom Controls, XAML
- **Frontend & UI:** React.js, Bootstrap, HTML5, CSS3, JavaScript
- **Data Science & AI:** Python (NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn), Jupyter Notebooks
- **Languages:** C#, Python, SQL
- **Databases & Messaging:** MSSQL, PostgreSQL, SQLite, MySQL, Influx DB, RabbitMQ
- **Cloud & DevOps:** Microsoft Azure, Docker, Git, GitHub, GitLab, CI/CD Pipelines, Jira, Postman
- **Architecture:** Clean Architecture, MVVM, CQRS, Repository Pattern, Unit Testing, Agile Methodologies

Professional Experience

Software Engineer – R&D | Aditya Infotech Ltd (CP Plus) *Mar 2026 – Present |*
Noida, India

Project 1: VMS (Video Management System) *UI Developer, Desktop Client (Admin & Operator Apps)*

- **Multi-App Desktop Architecture:** Built two role-based desktop applications (Admin + Operator) in Avalonia UI (.NET 10) using MVVM pattern with CommunityToolkit source generators, shared UI control library, and DI-based service composition.
- **Live Camera Video Wall:** Developed an infinite-scroll multi-camera video wall canvas with drag-and-drop camera placement, resizable/movable grid cells, fullscreen mode, and per-cell status overlays (connecting, online, error, offline) with debug diagnostics.
- **Camera & Recording Management UI**: Designed tabbed Add Camera dialog, recording profile editor, per-stream (main/sub) continuous recording toggles, and a weekly 24×7 schedule grid editor.
- **Role-Based Access & User Management:** Built full CRUD UI for users and roles with permission-key–based RBAC — Admin app gates access on RequiredPermission at login, auto-navigating or locking out based on the user's assigned role.
- **Tech Stack:** C#, Avalonia UI, .NET 10, MVVM, SignalR, JWT authentication

Project 2: PTS Tool (Production Testing System) *UI Developer for hardware device production workflow automation.*

- **Multi-Step Production Workflow UI:** Built a structured desktop workflow in Avalonia (.NET) covering Scan → Flash → Burn → Functional Test → Boot Test → Print Label — ensuring each step completes before the next unlocks, reducing operator errors on the production line.
- **Real-Time Device Discovery:** Implemented a network camera scanning panel using background threading and ObservableCollection, displaying discovered devices live on the UI as they are found without blocking the main thread.
- **Firmware Flashing with Progress Tracking:** Designed an interactive Flash panel with a real-time animated progress bar, firmware file verification (MD5 check), and cancellable async flash operations using CancellationTokenSource.
- **Serial & MAC Burn + Label Printing:** Built a Burn panel to write Serial Number and MAC Address to devices, with a label preview dialog and direct printer integration for physical label output.
- **Role-Based Navigation**: Integrated permission-based routing on startup — the app auto-navigates to the first permitted section (Dumping, SMT Testing, Tuning, Testing, Upgrade) based on the logged-in user's role fetched via JWT-authenticated API.
- **Theme & Language Support:** Developed a dynamic sidebar and topbar with collapsible navigation, multi-theme switching, and multi-language (i18n) support using a centralized AppSettings system.
- **Tech Stack:** C#, Avalonia UI, .NET, MVVM pattern, async/await, JWT authentication

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Software Engineer | Ajeevi Technologies Pvt Ltd *Mar 2024 – Feb 2026 | Noida, India*

Project 1: ICCC-Smart City (Integrated Command & Control Center) *Lead Backend Developer for Smart City IoT monitoring solution.*

- **High-Scale IoT Data:** Engineered real-time GPS dashboards and backend modules to process live telemetry data from 1,000+ IoT devices, ensuring sub-second reporting latency.
- **Modern API Architecture:** Designed and deployed high-performance backend services using .NET Core and Minimal APIs, reducing boilerplate code and improving endpoint execution speed.
- **Multi-Database Orchestration:** Integrated a hybrid database strategy using MSSQL/MySQL for relational data and Influx DB for high-velocity time-series sensor data.
- **Optimization:** Developed a library of reusable backend components, decreasing development time for new features by approximately 20%.

Project 2: Video Management System (VMS) *Backend API Specialist for video data processing.*

- **Architectural Implementation:** Leveraged the CQRS pattern to separate read/write operations, significantly improving API responsiveness and data consistency.
- **Performance Tuning:** Conducted deep-level database optimization in MSSQL, reducing query execution times for large video metadata sets.

Project 3: Smart Parking Management System (SPMS) *Backend Developer for real-time urban parking solutions.*

- **Scalable Service Design:** Built RESTful APIs and modularized backend services to allow for horizontal scaling as parking site density increases.
- **Real-time Synchronization:** Optimized MySQL and Influx DB queries to handle simultaneous parking occupancy updates from hundreds of concurrent sensors.
- **Reliability:** Refined API error-handling and debugging protocols, resulting in a more robust and fault-tolerant video processing pipeline.

Software Engineer Intern | So Infotech Pvt Ltd *Sep 2023 – Jeb 2024 | Noida, India*

Project: Concert-eTicket-Booking (Enterprise Ticket Management System)

- **End-to-End Development:** Engineered the core ticket-booking engine using ASP.NET Core MVC and SQL Server, focusing on transaction integrity during peak booking windows.
- **API Integration:** Developed secure REST APIs and user authentication modules.

Education

- **Master of Computer Applications (MCA)** *Mangalayatan University, Aligarh, UP* | Jan 2024 – Jan 2026

Certificate

- **Microsoft Certified: Azure Developer Associate** *Microsoft* | Issued 2026

Place: Delhi NCR

Sincerely,
Sushil Thakur